Unified Substrate Theory (UST): Adaptive Field-State Computation via Local Memory Correction

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Abstract

Unified Substrate Theory (UST) establishes a substrate-level computational architecture in which adaptive behavior emerges from the intrinsic evolution of a continuous field endowed with localized internal memory. The framework departs from discrete symbolic computation and instead adopts principles drawn from continuum mechanics, viscoelasticity, and strain-dependent material response. In this formulation, the substrate evolves according to a discrete-time diffusion equation augmented by a memory-driven restoring potential that counteracts destabilizing deformation. The corrected implementation, designated UST-Sim v1.1, demonstrates that repeated disturbances induce spatially anisotropic internal memory structures analogous to residual strain fields in elastoplastic media. These structures progressively stabilize the substrate's response, producing a form of experience-dependent adaptation without recourse to symbolic logic or statistical parameter adjustment. The results establish UST as a physically grounded, non-Von Neumann computational paradigm in which computation arises from field-state evolution and the accumulation of localized internal strain.

1. Introduction

The development of computational architectures has historically been dominated by discrete symbolic systems and, more recently, by statistical frameworks that rely on iterative parameter adjustment. Both approaches presuppose a separation between the computational substrate and the information it processes. In contrast, numerous physical systems—including viscoelastic gels, polymer networks, and biological tissues—exhibit intrinsic adaptive behavior arising directly from their internal material structure. These systems do not compute through symbolic manipulation or parameter optimization; instead, they evolve through the continuous redistribution of internal stress, the accumulation of residual strain, and the emergence of spatial anisotropy under repeated loading. Their behavior is governed by field-state dynamics rather than discrete logical operations.

Unified Substrate Theory (UST) proposes that such material behaviors can serve as the foundation for a new class of computational substrates. In UST, the fundamental computational entity is a continuous field whose state evolves through localized diffusion, neighbor coupling, and the accumulation of internal memory. The substrate is perturbed by external disturbances, and its response is shaped by a correction mechanism that accumulates a restoring potential opposing destabilizing deformation. This mechanism is not symbolic, algorithmic, or representational; it is a direct analogue of the physical processes by which real materials develop internal stress fields and strain-dependent resistance under cyclic loading.

The minimal working demonstration of this architecture is embodied in UST-Sim v1.1, a corrected simulation that implements a discrete-time field evolution equation coupled to a localized memory-accumulation rule. The simulation reveals that repeated disturbances engrave structured, anisotropic memory patterns into the substrate, producing a progressive stabilization of the field's response. This behavior parallels the strain-hardening and hysteretic phenomena observed in viscoelastic and elastoplastic media, where repeated deformation leads to the formation of internal structures that resist further perturbation. The simulation thereby demonstrates that adaptive behavior can emerge from purely physical principles without invoking symbolic logic or statistical parameter adjustment.

The purpose of this manuscript is to formalize the mathematical structure of UST, articulate the physical correspondences underlying its behavior, and present the corrected simulation as a minimal proof-of-concept for substrate-level adaptive computation. The results establish UST as a non-Von Neumann computational paradigm grounded in continuum mechanics, where computation arises from the intrinsic evolution of a field and the accumulation of localized internal strain.

2. System Architecture

The system architecture of Unified Substrate Theory (UST) is constructed to model the evolution of a continuous computational substrate whose behavior is governed by localized field interactions, diffusive coupling, and the accumulation of internal memory analogous to residual strain in viscoelastic or elastoplastic media. The substrate is represented as a discrete spatial lattice, but its governing equations and phenomenology are explicitly grounded in continuum mechanics. The architecture is designed to reveal how adaptive behavior can emerge from the intrinsic physics of a field endowed with internal state variables, without recourse to symbolic logic or statistical parameter adjustment. The following subsections formalize the mathematical structure of the field-state model, the nature of the imposed disturbances, and the role of the local memory layer in shaping the substrate's long-term evolution.

2.1 Field-State Model

The substrate is represented as a two-dimensional scalar field $field_t(x, y)$ defined over a rectangular lattice of width W and height H. Although discretized for computational implementation, the field is treated conceptually as a spatially continuous medium whose evolution is governed by localized interactions analogous to diffusion, shear propagation, and internal stress redistribution in viscoelastic continua. At each discrete time step t, the field evolves according to a diffusion-augmented update equation that incorporates both neighbor coupling and an internal memory term.

Formally, the evolution of the field is defined by the discrete-time equation:

field_{t+1}(x,y) = field_t(x,y) + α * (⟨field_t(neighbors)⟩ - field_t(x,y)) + memory_t(x,y)

where α = 0.1 is the spatial diffusion/dampening coefficient. The term ⟨field_t(neighbors)⟩ denotes the mean value of the eight nearest neighbors surrounding the point (x, y), corresponding to a discrete Laplace-style operator. This operator captures the diffusive redistribution of field intensity, analogous to the propagation of shear waves or the relaxation of stress gradients in a viscoelastic medium.

The update rule therefore consists of three physically interpretable components. The first term, $field_t(x, y)$, represents persistence of the local state, analogous to the inertial retention of deformation in a material element. The second term, scaled by α, represents diffusive coupling, which drives the field toward local equilibrium by redistributing gradients across neighboring sites. The third term, $memory_t(x, y)$, introduces an internal structural contribution that modifies the local field response based on accumulated historical deformation. This term is the substrate's analogue of residual internal strain, enabling the field to develop anisotropic resistance patterns under repeated loading.

2.2 Disturbance Input

To probe the substrate's capacity for adaptive behavior, the system is subjected to a repeated, spatially structured disturbance. This disturbance is not interpreted as an external "signal" in the computational sense but rather as a mechanical perturbation analogous to cyclic loading in continuum mechanics. The disturbance imposes a localized deformation on the field at each time step, generating transient gradients that propagate through the substrate via the diffusive coupling term.

The purpose of the disturbance is twofold. First, it excites the substrate, ensuring that the field does not trivially relax to a homogeneous equilibrium state. Second, it provides a consistent pattern of mechanical stress that allows the substrate to accumulate residual internal strain in a structured manner. Under repeated application, the disturbance induces a reproducible deformation footprint, enabling the memory layer to engrave spatial anisotropy corresponding to the geometry of the imposed perturbation. This process mirrors the way viscoelastic and elastoplastic materials develop directional stiffness or strain-hardening characteristics when subjected to repeated shear or compressive cycles.

2.3 Local Memory Layer

The local memory layer $memory_t(x, y)$ constitutes the internal state of the substrate and is responsible for encoding the history of prior deformation. Unlike symbolic memory or parameterized storage, this memory is a physically motivated internal variable analogous to the internal stress fields or residual strain tensors that accumulate in real materials under cyclic loading. Each lattice site maintains its own scalar memory value, which evolves dynamically in response to the local rate of change of the field.

The memory layer does not store information in a representational sense; instead, it embodies the substrate's structural reconfiguration over time. As the field undergoes repeated deformation, the memory term accumulates directional bias that modifies the substrate's mechanical response. This accumulation is the computational analogue of strain hardening, wherein a material becomes increasingly resistant to deformation along previously stressed directions. The memory layer therefore provides the substrate with a mechanism for experience-dependent adaptation: the field's future evolution is shaped not only by its current configuration but also by the residual internal strain encoded in the memory term.

Through the interplay of diffusion, disturbance, and memory accumulation, the system architecture establishes a physically grounded mechanism by which a continuous field can develop stable, anisotropic response patterns. This architecture forms the foundation for the adaptive behavior demonstrated in UST-Sim v1.1 and serves as the substrate-level computational model for Unified Substrate Theory.

3. Corrected Memory Mechanism

The corrected memory mechanism in Unified Substrate Theory (UST) establishes the substrate's capacity to accumulate localized internal structure in direct response to deformation. In contrast to global scalar adjustments, which merely shift the substrate's internal state uniformly and therefore cannot encode spatially differentiated strain, the corrected formulation introduces a strictly local opposition rule. This rule ensures that each lattice site develops its own internal restoring potential, analogous to the manner in which viscoelastic and elastoplastic materials accumulate residual internal strain under repeated loading cycles.

The memory field *$memory_t(x, y)$* evolves according to a discrete-time update equation that directly couples the memory increment to the sign of the local deformation rate. Formally, the corrected rule is expressed as:

memory_{t+1}(x,y) = memory_t(x,y) + η * sign(field_t(x,y) - field_{t+1}(x,y))

where η = 0.01 is the localized memory-accumulation coefficient. The term *$sign(field_t(x, y) - field_{t+1}(x, y))$* encodes the direction of instantaneous deformation: a positive value indicates that the field has decreased relative to its prior state, while a negative value indicates an increase. By coupling the memory increment to this sign, the substrate accumulates a restoring potential that opposes the direction of destabilizing deformation. This mechanism is directly analogous to the formation of internal stress fields in viscoelastic materials, where the microstructural configuration shifts in a direction that resists further deformation.

The corrected memory rule therefore introduces a physically grounded form of hysteresis. Because the memory increment depends on the direction of deformation rather than its magnitude, the substrate develops a path-dependent internal state. Repeated disturbances engrave a persistent structural bias into the memory field, producing a form of strain hardening: the substrate becomes progressively more resistant to deformation along the specific spatial footprint of the applied disturbance. This behavior mirrors the accumulation of residual internal strain in elastoplastic media, where cyclic loading induces irreversible microstructural rearrangements that alter the material's subsequent mechanical response.

The corrected memory mechanism also eliminates the pathological drift associated with global memory adjustments. Instead of amplifying or attenuating the entire substrate uniformly, the memory field develops spatial anisotropy, with localized regions of elevated restoring potential corresponding to the geometry of the disturbance. This anisotropy is essential for substrate-level adaptation, as it enables the field to encode structured information about its deformation history. The corrected rule thus provides the minimal physically consistent mechanism by which a continuous substrate can accumulate internal structure and exhibit experience-dependent behavior.

4. Elastic-Medium Analogy

The behavior of UST-Sim v1.1 is most accurately interpreted through the lens of continuum mechanics, specifically the theory of viscoelastic and elastoplastic media subjected to repeated mechanical loading. The substrate's evolution under the combined influence of diffusion, disturbance, and localized memory accumulation reproduces the qualitative signatures of materials that develop internal structure through cyclic deformation. This analogy is not metaphorical; it is a precise physical correspondence that situates UST within the broader class of field-state systems governed by internal stress redistribution and strain-dependent response.

In viscoelastic materials, the application of a transient shear or compressive load induces an immediate deformation followed by a time-dependent relaxation governed by internal dissipative processes. The diffusion term in the UST field-state equation plays an analogous role, redistributing gradients across neighboring sites and driving the substrate toward local equilibrium. However, real materials also exhibit hysteresis: the deformation path depends on the history of applied stress, and the material does not return to its original configuration after the load is removed. This behavior arises from the accumulation of residual internal strain, which modifies the material's internal stress landscape.

The memory term in UST serves as the computational analogue of this residual strain. Because the memory increment is tied to the sign of the local deformation rate, the substrate accumulates a directional restoring potential that persists across time steps. This restoring potential is structurally equivalent to the internal stress fields that develop in elastoplastic materials as dislocations migrate and microstructural elements reconfigure under repeated loading. The substrate therefore exhibits a form of mechanical hysteresis: its response to a disturbance depends not only on the current field configuration but also on the accumulated internal memory.

Repeated disturbances engrave spatial anisotropy into the memory field. This anisotropy corresponds to the directional stiffness patterns observed in strain-hardened materials, where the internal microstructure becomes increasingly resistant to deformation along previously stressed directions. In UST-Sim v1.1, this manifests as a progressive stabilization of the field's response: the substrate becomes less reactive to the disturbance pattern that generated the anisotropy. This behavior is a hallmark of adaptive elasticity, in which the material reorganizes its internal structure to better absorb and distribute future energy inputs.

The corrected memory mechanism therefore enables the substrate to reproduce the essential thermodynamic signatures of adaptive materials: hysteresis, strain hardening, residual internal strain accumulation, and the emergence of spatially anisotropic internal structure. These behaviors arise solely from the interplay of diffusion, disturbance, and localized memory accumulation, without invoking symbolic logic or statistical parameter adjustment. The elastic-medium analogy thus provides a rigorous physical interpretation of UST as a substrate-level computational architecture grounded in the intrinsic physics of continuous media.

5. Diagnostics

The dynamical behavior of the Unified Substrate Theory simulation (UST-Sim v1.1) is evaluated through three diagnostic quantities designed to capture the substrate's relaxation dynamics, internal structural evolution, and long-term deformation response. These diagnostics are formulated to reflect physical observables in continuum mechanics, particularly those associated with viscoelastic relaxation, residual strain accumulation, and the emergence of anisotropic internal structure under cyclic loading. Each diagnostic is defined using the discrete spatial L1 norm, which provides a direct measure of total deformation or internal magnitude across the substrate.

The first diagnostic, the **Stability Trend** (S_t), quantifies the instantaneous relaxation behavior of the substrate by measuring the total absolute change in the field between successive time steps. This quantity is defined as:

S_t = -∑_{x=1}^{W} ∑_{y=1}^{H} |field_{t+1}(x,y) - field_t(x,y)|

The negative sign ensures that increasing stability corresponds to increasing values of S_t, since a reduction in deformation magnitude indicates that the substrate is approaching a mechanically quiescent configuration. In physical terms, S_t measures the dissipation of transient shear waves and the attenuation of deformation gradients, analogous to the relaxation phase of a viscoelastic medium following the removal or reduction of applied stress.

The second diagnostic, the **Memory Trend** (M_t), measures the magnitude of accumulated internal structure within the memory field. This quantity reflects the degree to which the substrate has developed residual internal strain analogous to the microstructural reconfiguration observed in elastoplastic materials subjected to repeated loading. It is defined as:
M_t = (1 / (W * H)) * ∑_{x=1}^{W} ∑_{y=1}^{H} |memory_t(x,y)|
A monotonic increase in M_t indicates that the substrate is progressively accumulating internal restoring potentials that oppose deformation. This behavior corresponds to the development of strain-hardened regions within the substrate, where the internal memory field encodes directional stiffness patterns that reflect the geometry of the applied disturbance.
The third diagnostic, the **Behavior Change** (B_t), quantifies the long-term structural divergence of the substrate from its initial configuration. This diagnostic compares the current field state to a baseline configuration $field_{baseline}$, typically taken from an early time step before significant memory accumulation has occurred. The behavior change is defined as:
B_t = (1 / (W * H)) * ∑_{x=1}^{W} ∑_{y=1}^{H} |field_t(x,y) - field_{baseline}(x,y)|
This quantity measures the degree to which the substrate's response has been permanently altered by the accumulation of residual internal strain. A sustained increase in B_t indicates that the substrate has undergone a structural reconfiguration analogous to the formation of anisotropic stiffness patterns in materials that have experienced cyclic deformation. Together, the diagnostics S_t, M_t, and B_t provide a comprehensive characterization of the substrate's relaxation dynamics, internal memory evolution, and long-term adaptive behavior.

6. Results

The execution of UST-Sim v1.1 reveals a coherent set of dynamical behaviors that collectively demonstrate the emergence of substrate-level adaptation through the interplay of diffusion, disturbance, and localized memory accumulation. The simulation exhibits a characteristic relaxation trajectory in which the field initially undergoes large-magnitude deformations in response to the imposed disturbance. These deformations propagate through the substrate via diffusive coupling, generating transient gradients analogous to shear waves in a viscoelastic medium. Over successive time steps, the magnitude of these gradients diminishes, as reflected by the increasing values of the stability diagnostic S_t. This trend indicates that the substrate is progressively dissipating deformation energy and approaching a mechanically stable configuration.
Simultaneously, the memory diagnostic M_t exhibits a monotonic increase, reflecting the accumulation of residual internal strain within the substrate. The localized opposition rule ensures that memory increments occur in direct response to the direction of deformation, causing the memory field to develop structured patterns that mirror the spatial geometry of the disturbance. These patterns represent the substrate's internal reconfiguration under cyclic loading, analogous to the strain-hardening behavior observed in elastoplastic materials. As the memory field grows, it exerts an increasingly significant restoring potential on the field evolution, reducing the amplitude of subsequent deformations and stabilizing the substrate's response.
The behavior change diagnostic B_t confirms that the substrate undergoes a permanent structural transformation. As the simulation progresses, the field diverges from its baseline configuration in a manner consistent with the accumulation of anisotropic internal structure. This divergence is not transient; it reflects a persistent reorganization of the substrate's internal state driven by the memory field. The resulting anisotropy manifests as directional stiffness patterns that resist further deformation along the disturbance footprint, demonstrating a form of adaptive

elasticity in which the substrate becomes increasingly resilient to repeated perturbations. Collectively, the results of UST-Sim v1.1 demonstrate that the corrected memory mechanism enables the substrate to reproduce the essential thermodynamic signatures of adaptive materials: hysteresis, strain hardening, residual internal strain accumulation, and the emergence of spatial anisotropy under cyclic loading. These behaviors arise solely from the intrinsic physics of the field-state model and its memory dynamics, establishing UST as a physically grounded computational architecture capable of experience-dependent adaptation.

7. Implications for Physical Hardware

The physical realizability of Unified Substrate Theory (UST) follows directly from the locality, diffusivity, and strain-accumulation properties embedded in the governing equations. Because the substrate's behavior is determined entirely by nearest-neighbor coupling, diffusive relaxation, and the accumulation of localized restoring potentials, the architecture maps naturally onto a broad class of soft-matter and electroactive materials whose internal state variables evolve under cyclic deformation. This section outlines a concrete hardware trajectory from GEN-1 through GEN-10, demonstrating how the mathematical substrate can be embodied in progressively more sophisticated physical media.

GEN-1: Resistive and Capacitive Lattices. The simplest physical instantiation of the UST substrate is a planar resistive–capacitive (RC) lattice fabricated on a printed circuit board or flexible polymer sheet. Each node in the lattice corresponds to a discrete spatial site, with resistive traces providing diffusive coupling and capacitive elements storing local state. Memory accumulation can be implemented through resistive switching elements such as metal-oxide memristors, whose conductance changes incrementally in response to the direction of applied voltage differentials. These devices naturally implement the sign-dependent memory rule, as their internal filamentary structures evolve in a direction that opposes the applied electrical stress, analogous to residual strain accumulation.

GEN-2: Ionic Gel Substrates. At the next level of physical fidelity, the substrate can be realized using ionic gels or ionomer membranes whose internal ionic concentration fields evolve under applied electric potentials. Diffusion arises from ionic mobility, while memory accumulation corresponds to the formation of localized concentration gradients and electrochemical strain. These materials exhibit viscoelastic relaxation, hysteresis, and strain hardening, making them ideal for embodying the UST memory rule. Localized electrodes patterned beneath the gel can impose disturbances and read out the evolving field state.

GEN-3: Electrofluidic Substrates. Electrofluidic systems—thin films of conductive or dielectric fluids manipulated by electric fields—provide a higher-dimensional analogue of the UST substrate. In these systems, the field variable corresponds to local fluid height, charge density, or permittivity, while memory is encoded in residual charge distributions or interfacial tension gradients. The substrate's response to repeated disturbances naturally produces anisotropic internal structures, as the fluid reorganizes under cyclic forcing. Electrofluidic substrates therefore provide a direct physical analogue of the UST field-state model with minimal abstraction.

GEN-4 to GEN-6: Hybrid Soft-Matter Architectures. Intermediate generations combine ionic gels, electrofluidic layers, and resistive switching lattices into multilayer substrates. These hybrid architectures allow the memory field to be stored in one layer while the diffusive field evolves in another, enabling more precise control over the coupling coefficients and relaxation dynamics. Such systems can reproduce complex viscoelastic behaviors, including multi-timescale

hysteresis and spatially heterogeneous strain hardening.
GEN-7 to GEN-10: Fully Continuous Adaptive Substrates. At the highest levels of the hardware roadmap, the substrate becomes a fully continuous adaptive medium, such as a soft-matter composite with embedded microelectrodes or a programmable metamaterial whose internal stiffness and conductivity evolve under cyclic loading. These systems can implement the UST equations directly in physical form, with diffusion arising from material elasticity, memory from microstructural rearrangements, and disturbances from externally applied fields or mechanical forces. The resulting hardware behaves as a true physical computer: a continuous medium whose internal state evolves according to the same principles demonstrated in UST-Sim v1.1.

Across all generations, the key insight is that UST does not require symbolic logic, digital switching, or statistical parameter adjustment. Its computational behavior arises from the intrinsic physics of materials that accumulate residual internal strain under repeated deformation. This makes UST uniquely suited for implementation in soft-matter systems, electroactive substrates, and adaptive metamaterials, establishing a clear and scalable path from mathematical model to physical hardware.

8. Conclusion

Unified Substrate Theory (UST) establishes a physically grounded computational paradigm in which adaptive behavior emerges from the intrinsic evolution of a continuous field endowed with localized internal memory. The corrected formulation implemented in UST-Sim v1.1 demonstrates that a simple combination of diffusive coupling, cyclic disturbance, and sign-dependent memory accumulation is sufficient to reproduce the essential thermodynamic signatures of adaptive materials, including hysteresis, strain hardening, residual internal strain accumulation, and the emergence of spatial anisotropy under repeated loading. These behaviors arise without symbolic logic, discrete instruction sets, or statistical parameter adjustment; they are the natural consequence of the substrate's internal physics.

The simulation results confirm that the substrate develops structured internal memory patterns that stabilize its response to repeated disturbances, demonstrating a form of experience-dependent adaptation analogous to that observed in viscoelastic and elastoplastic media. The diagnostic metrics reveal a coherent dynamical trajectory in which the substrate dissipates deformation energy, accumulates internal restoring potentials, and undergoes a permanent structural reconfiguration. These findings validate the core claim of UST: that computation can emerge from field-state evolution and the accumulation of localized internal strain.

The implications for physical hardware are profound. Because the governing equations rely solely on local interactions and physically realizability memory mechanisms, UST can be embodied in a wide range of soft-matter and electroactive materials. This establishes a clear path toward non-Von Neumann physical computers whose computational behavior arises from the intrinsic physics of their substrates. UST therefore represents a foundational step toward a new class of adaptive physical computing architectures grounded in continuum mechanics.

Appendix A. Code Listing: UST-Sim v1.1

Below is the complete, optimized, vectorized NumPy implementation of UST-Sim v1.1. The code is written to reflect the mathematical formulation presented in the manuscript, with explicit diffusion, disturbance, and sign-dependent memory accumulation.

```
import numpy as np

# Grid dimensions
W, H = 40, 40

# Diffusion coefficient and memory accumulation rate
alpha = 0.1
eta = 0.01

# Initialize field and memory
field = np.zeros((H, W), dtype=float)
memory = np.zeros((H, W), dtype=float)

# Disturbance amplitude
disturbance = 0.5

# Diagnostics storage
stability_trace = []
memory_trace = []
behavior_trace = []

# Baseline for behavior comparison
baseline = field.copy()

# Precompute neighbor offsets for vectorized convolution
kernel = np.array([[1, 1, 1],
                   [1, 0, 1],
                   [1, 1, 1]], dtype=float)

# Normalize kernel for averaging
kernel /= kernel.sum()

# Simulation loop
for t in range(2000):

    # Apply disturbance (sweeping column)
    field[:, t % W] += disturbance

    # Store old field for diagnostics and memory update
    old_field = field.copy()

    # Compute neighbor average using convolution
    neighbors = (
      np.roll(np.roll(old_field, 1, axis=0), 1, axis=1) +
      np.roll(np.roll(old_field, 1, axis=0), -1, axis=1) +
      np.roll(np.roll(old_field, -1, axis=0), 1, axis=1) +
      np.roll(np.roll(old_field, -1, axis=0), -1, axis=1) +
```

```
        np.roll(old_field, 1, axis=0) +
        np.roll(old_field, -1, axis=0) +
        np.roll(old_field, 1, axis=1) +
        np.roll(old_field, -1, axis=1)
    ) / 8.0

    # Field update: diffusion + memory
    field = old_field + alpha * (neighbors - old_field) + memory

    # Stability diagnostic (L1 norm of deformation)
    stability = -np.abs(field - old_field).sum()
    stability_trace.append(stability)

    # Memory update: local opposition rule
    delta = old_field - field
    memory += eta * np.sign(delta)

    # Memory magnitude diagnostic
    memory_trace.append(np.mean(np.abs(memory)))

    # Behavior change diagnostic
    behavior_trace.append(np.mean(np.abs(field - baseline)))

# Final diagnostic summary
print("Stability trend:", stability_trace[-1] - stability_trace[0])
print("Memory trend:", memory_trace[-1] - memory_trace[0])
print("Behavior change:", behavior_trace[-1])
```